higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIAL TECHNOLOGY
NQF LEVEL 3

14 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
1.1.1 B
1.1.2 B
1.1.3 C
1.1.4 C
1.1.5 A

(5 × 1) (5)

1.2
1.2.1 The density of a material is its mass✓ per unit volume.✓

1.2.2 The melting point of a material is the temperature✓ at which the material changes from a solid state to a liquid state.✓

1.2.3 The thermal conductivity of a material is the rate at which heat✓ can flow through the material.✓

(3 × 2) (6)

1.3
1.3.1 The hardness of the steel will increase.

1.3.2 The tensile strength of the steel will increase.

1.3.3 The ductility of the steel will decrease.

1.3.4 It will become more difficult to machine the steel as the carbon content of the steel increases.

(4 × 1) (4)

1.4
1.4.1 The charge is heated by the reflection✓ of the heat from the fire off the slant-shaped furnace roof.✓ (2)

1.4.2
- Pulverised coal
- Liquid petroleum (2)

1.4.3 The firebricks can withstand very high temperatures, have low thermal conductivity,✓ and act as insulation against the loss of heat through the furnace walls,✓ ensuring greater energy efficiency.✓ (3)

1.5
- Increases the tensile strength of the steel
- Hardens the steel
- Increases the steel's resistance to heat
- Prevents brittleness when the steel is tempered
- Increases the depth of hardness of the steel (Any 3 × 1) (3)

[25]

QUESTION 2

2.1	2.1.1	True		
	2.1.2	False		
	2.1.3	True		
	2.1.4	False		
	2.1.5	False		
	2.1.6	True		
	2.1.7	True		
	2.1.8	True		
	2.1.9	True		
	2.1.10	False		
			(10 × 1)	(10)
2.2	2.2.1	• High strength • Durability • Light in weight • Low density • Good insulation • Corrosion resistance	(Any 4 × 1)	(4)
	2.2.2	• Headlights • Side or door panels • Bumpers • Grill • Instrument panel • Spoiler	(Any 4 × 1)	(4)
2.3	2.3.1	• Elasticity • Durability • Electrical insulation • Thermal insulation • Flexibility	(Any 4 × 1)	(4)
	2.3.2	• Ethylene-propylene rubbers • Acryl elastomers • Silicon rubbers • Butyl rubbers	(Any 3 × 1)	(3)
				[25]

QUESTION 3

3.1
3.1.1 K
3.1.2 H
3.1.3 J
3.1.4 G
3.1.5 I
3.1.6 B
3.1.7 F
3.1.8 C
3.1.9 D
3.1.10 E

(10 × 1) (10)

3.2 3.2.1 The bottom of the furnace is covered with lime✓ and then charged with scrap metal.✓ (2)

3.2.2 The furnace is tilted slightly✓ and the slag is manually raked off into a ladle.✓ (2)

3.2.3 The temperature of the molten steel is checked,✓ and, if correct, the steel is tapped into the ladle.✓ Ingot moulds are filled from the ladle.✓ (3)

3.3 3.3.1
- Annealing
- Normalising
- Carburising
- Hardening (Any 3 × 1) (3)

3.3.2
- Heating work piece to specified temperature
- Hold temperature for required time in process called soaking
- Slowly cool down work piece over determined time (3)

3.3.3
- Cost is very high✓
- Ageing temperature is critical✓
- Time is critical✓ (Any 2 × 1) (2)

[25]

QUESTION 4

4.1
4.1.1 indenter
4.1.2 dwell
4.1.3 10 to 15
4.1.4 optically
4.1.5 dividing

(5 × 1) (5)

4.2 Stress (σ) : Pascal✓ OR N/m^2✓
Force(F) : Newton✓ OR N✓
Area(A) : square metres✓ OR m^2✓ (3)

4.3 4.3.1

$$\frac{F}{D^2} = 30$$
$$F = 30.D^2$$
$$F = 30 \times 10^2 \checkmark$$
$$\therefore F = 3000\ kg \checkmark$$

(2)

4.3.2

$$Surface\ area\ of\ impression = \frac{\pi D}{2}\left\{D - \sqrt{D^2 - d^2}\right\}$$
$$= \frac{\pi \times 10}{2}\left\{10 - \sqrt{10^2 - 4.17^2}\right\} \quad (\checkmark \text{ and } \checkmark)$$
$$= 14{,}309\ mm^2 \checkmark$$

Accurate to three decimal places✓ (4)

4.3.3

$$BHN = \frac{load\ kg}{curved\ surface\ area\ of\ indentation\ (mm^2)}$$
$$= \frac{3000}{14{,}309} \checkmark$$
$$= 209{,}658 \checkmark$$

(3)

Accurate to three decimal places✓

4.3.4

$$Tensile\ strength = 3{,}24 \times BHN$$
$$= 3{,}24 \times 209{,}658$$
$$= 679{,}29 MPa \checkmark$$

Accurate to two decimal places✓ (2)

4.4

$$\sigma = \frac{F}{A}$$
$$A = \frac{F}{\sigma} = \frac{16000}{47{,}23 \times 10^6} \checkmark$$
$$\therefore A = 338{,}7677 \times 10^{-6} m^2 \checkmark$$

$$A = \frac{\pi}{4} D^2$$
$$D^2 = \frac{4A}{\pi} \checkmark$$
$$D = \sqrt{\frac{4 \times 338{,}7677 \times 10^{-6}}{\pi}} \checkmark$$
$$\therefore D = 0{,}020769 m = 20{,}77 mm \checkmark$$

Accurate to two decimal places✓

(6)
[25]

TOTAL: 100

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